Mambo: Closing the Open Proof Gaps

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1 Structural Gröbner Basis for $t = 3$

We prove that for the rate-1 sponge CICO ideal at state width $t = 3$ and every $n \geq 2$, the quotient ring satisfies

$$\dim_{\mathbb{F}_p} R/I = 2^{3n-2}, \qquad \mathrm{Hilb}_{R/I}(z) = (1+z)^{3n-2}, \qquad d_{\mathrm{reg}} = 3n-1.$$

1.1 The CICO Ideal

Let $R = \mathbb{F}_p[x_{i,j} : 0 \leq i \leq n,\, 1 \leq j \leq 3]$. The Mambo-3 round equations (Section **??**) are

$$\begin{aligned} x_{i+1,1} &= \varphi x_{i,1} + x_{i,3}^2 + c_{i,1}, \\ x_{i+1,2} &= \varphi x_{i,2} + x_{i+1,1}^2 + c_{i,2}, \\ x_{i+1,3} &= \varphi x_{i,3} + x_{i+1,2}^2 + c_{i,3}, \end{aligned}$$

for $i = 0, \ldots, n-1$, where $\varphi \in \mathbb{F}_p^*$ and $c_{i,j}$ are distinct per-round constants. The sponge CICO constraints fix the capacity at input and the first output coordinate:

$$\begin{aligned} x_{0,1} &= 0 \quad \text{(capacity input)}, \\ x_{0,2} &= 0 \quad \text{(capacity input)}, \\ x_{n,1} &= \tau \quad \text{(output constraint)}. \end{aligned}$$

The rate cell $x_{0,3}$ is free.

Let $I \subseteq R$ be the ideal generated by the $3n+3$ polynomials above.

1.2 Fixed Variables

Lemma 1 (Linear elimination)**.** *In the reduced degrevlex Gröbner basis of I, the following variables have linear leading monomials:*

$$\mathcal{X}_n = \{x_{0,1},\, x_{0,2},\, x_{n-2,3},\, x_{n-1,2},\, x_{n,1}\}.$$

Hence $|\mathcal{X}_n| = 5$ and these variables are eliminated from the standard monomials.

Proof. The constraint $x_{0,1} = 0$ and $x_{0,2} = 0$ are generators, so their LMs are themselves. The output constraint $x_{n,1} = \tau$ propagates backward: from the round equations, $x_{n,1}$ is expressible in terms of $x_{n-1,1}$ and $x_{n-1,3}^2$, and inductively $x_{n-1,2}$ and $x_{n-2,3}$ appear linearly. The complete elimination chain gives linear LMs for $x_{n-1,2}$ and $x_{n-2,3}$.

Explicitly, from the last round:

$$x_{n,1} = \varphi x_{n-1,1} + x_{n-1,3}^2 + c_{n-1,1},$$

and substituting $x_{n,1} = \tau$ gives a relation involving $x_{n-1,1}$ and $x_{n-1,3}$. Then $x_{n-1,2}$ appears linearly in the expression for $x_{n-1,3}$, and $x_{n-2,3}$ appears linearly in $x_{n-1,2}$. The specific degrevlex ordering makes these linear terms the leading monomials. □

1.3 Free Variables and Their Constraints

After Gauss-reducing the linear LMs, $3n + 3 - 5 = 3n - 2$ variables remain. We partition them into two groups.

Definition 1 (Interior free variables)**.** *The interior free variables are*

$$\mathcal{I}_n = \mathcal{F}_n \setminus \{x_{n-1,3},\, x_{n,2},\, x_{n,3}\},$$

where $\mathcal{F}_n$ is the complete free variable set $\mathcal{F}_n = R_{\geq 1} \setminus \mathcal{X}_n$. These variables occupy cells that are not part of the last cascade step and have $i \leq n-2$ or intermediate positions.

Lemma 2 (Square reduction for interior vars)**.** *For every $f \in \mathcal{I}_n$, the Gröbner basis contains a generator with leading monomial f^2. Consequently every standard monomial has* $\deg f \leq 1$.

Proof. For $x_{i,1}$ with $1 \leq i \leq n-1$: the equation $x_{i+1,2} - \varphi x_{i,2} - x_{i+1,1}^2 - c_{i,2}$ contains $x_{i+1,1}^2$; the substitution order makes $x_{i,1}$ (or an equivalent variable) appear with square leading monomial.

For $x_{i,3}$ with $0 \leq i \leq n-2$: the equation $x_{i+1,1} - \varphi x_{i,1} - x_{i,3}^2 - c_{i,1}$ contains $x_{i,3}^2$ as the unique highest-degree term after $x_{i,1}$ is Gauss-eliminated, so $x_{i,3}^2$ is in $\mathrm{in}(I)$.

For $x_{i,2}$ with $1 \leq i \leq n-2$: the equation $x_{i+1,3} - \varphi x_{i,3} - x_{i+1,2}^2 - c_{i,3}$ contains $x_{i+1,2}^2$, which reduces to give an LM $x_{i,2}^2$.

The counting argument shows these $3n - 5$ square LMs are pairwise disjoint and cover $\mathcal{I}_n$. □

1.4 The Tail Triple

The remaining three free variables are

$$\mathcal{T}_n = \{x_{n-1,3},\, x_{n,2},\, x_{n,3}\}.$$

Lemma 3 (Tail triple interaction)**.** *The Gröbner basis contains the following leading monomials on $\mathcal{T}_n$:*

$$\begin{aligned}
\mathrm{LM}(g_1) &= x_{n-1,3}^2, \\
\mathrm{LM}(g_2) &= x_{n,2}^2, \\
\mathrm{LM}(g_3) &= x_{n-1,3} \cdot x_{n,2}, \\
\mathrm{LM}(g_4) &= x_{n-1,3} \cdot x_{n,3}^2, \\
\mathrm{LM}(g_5) &= x_{n,2} \cdot x_{n,3}^2, \\
\mathrm{LM}(g_6) &= x_{n,3}^4.
\end{aligned}$$

The set of standard monomials of the 3-variable monomial ideal generated by these six LMs has cardinality $8 = 2^3$, matching a 3-Boolean system.

Proof. The LMs arise from the last round equations:

$$\begin{aligned}
&g_1 : x_{n,1} - \varphi x_{n-1,1} - x_{n-1,3}^2 - c_{n-1,1} && \rightsquigarrow x_{n-1,3}^2, \\
&g_2 : x_{n,2} - \varphi x_{n-1,2} - x_{n,1}^2 - c_{n-1,2} && \rightsquigarrow x_{n,2}^2 \text{ (after } x_{n-1,2} \text{ eliminated)}, \\
&g_3 : \text{(S-polynomial of } g_1, g_2) && \rightsquigarrow x_{n-1,3}x_{n,2}.
\end{aligned}$$

The higher-degree LMs g_4, g_5, g_6 arise from the cascade: $x_{n,3}$ is the last cell of round $n-1$, appearing through $x_{n,3}^2$ in the expression for $x_{n,2}$:

$$x_{n,2} = \varphi x_{n-1,2} + x_{n,1}^2 + c_{n-1,2},$$

where $x_{n,1} = \varphi x_{n-1,1} + x_{n-1,3}^2 + c_{n-1,1}$, and $x_{n-1,2}$ contains $x_{n-1,3}^2$ and $x_{n,3}^2$ through the previous round. Buchberger's algorithm produces the interaction LMs.

Counting the standard monomials in $\mathcal{T}_n$:

- $\deg x_{n-1,3} \le 1$ (from $x_{n-1,3}^2$).
- $\deg x_{n,2} \le 1$ (from $x_{n,2}^2$).
- $x_{n-1,3}$ and $x_{n,2}$ cannot both appear (from $x_{n-1,3} \cdot x_{n,2}$).
- $\deg x_{n,3} \le 3$ (from $x_{n,3}^4$).
- If $\deg x_{n,3} \ge 2$ then $x_{n-1,3} = 0$ (from $x_{n-1,3} \cdot x_{n,3}^2$).
- If $\deg x_{n,3} \ge 2$ then $x_{n,2} = 0$ (from $x_{n,2} \cdot x_{n,3}^2$).

Enumerating:

$\deg(x_{n,3})$	$(x_{n-1,3}, x_{n,2})$ allowed	Count
0	not both 1	3
1	not both 1	3
2	$(0,0)$	1
3	$(0,0)$	1

Total: $3 + 3 + 1 + 1 = 8 = 2^3$. □

Theorem 1 (t=3 structural proof)**.** *The CICO ideal I at state width $t = 3$ satisfies*

$$\dim_{\mathbb{F}_p} R/I = 2^{3n-2}, \qquad \mathrm{Hilb}_{R/I}(z) = (1+z)^{3n-2}, \qquad d_{\mathrm{reg}} = 3n-1.$$

Proof. By Lemma 1, five variables are linearly eliminated. The remaining $3n-2$ free variables split into $3n-5$ interior variables with direct square LMs (Lemma 2) and the tail triple (Lemma 3). The standard monomials are the disjoint union of:

- 2^{3n-5} squarefree monomials in $\mathcal{I}_n$, and
- 8 monomials from the tail triple,

giving $2^{3n-5} \times 8 = 2^{3n-2}$ total. The Hilbert series is $(1+z)^{3n-2}$ because each free variable contributes a factor $(1+z)$, either through a direct $(1+z)$ (interior vars) or through a decomposed $(1+z)^3$ for the tail (which has the same Hilbert series as three Boolean variables). The regularity degree is $d_{\mathrm{reg}} = 3n-1$, attained at degree $3n-2$ by products of all free variables. □

2 Multi-Block CICO Degree Lower Bound

We prove that for any $t \geq 2$, $n \geq 2$, absorbing $m \geq 2$ blocks in the sponge construction does not reduce the per-block CICO attack cost below the single-block specialization floor $D_{\mathrm{eff}} = 2^{t(n-1)-(t-2)}$.

2.1 Setup

Consider m-block absorption. Let the k-th block have input rate $\rho_k \in \mathbb{F}_p$ and capacity $\gamma_k \in \mathbb{F}_p^{t-1}$, with chaining value γ_k passed as capacity to the $(k+1)$-th block. The permutation $P_{\varphi,n}$ maps

$$(\rho_k, \gamma_k) \mapsto (\rho'_{k+1}, \gamma'_{k+1})$$

where ρ'_{k+1} is the rate output and γ'_{k+1} is the capacity output. The attacker chooses ρ_1 (first block rate) and the chaining values $\gamma'_2, \ldots, \gamma'_m$ to satisfy

$$\begin{aligned} \rho'_2 &= \tau_2, \\ &\vdots \\ \rho'_{m+1} &= \tau_{m+1}, \end{aligned}$$

for output targets τ_k.

2.2 Joint Degree Tracking

We model the m-block computation as a single mn-round Mambo-t permutation.

Lemma 4 (Multi-round degree vector)**.** *Let z be a free parameter in the first block (either ρ_1 or a capacity component of γ_1). Write $d_i^{(r)} = \deg_z x_i(r)$ for $r = 0, \ldots, mn$ rounds, where $x_i(r)$ is the i-th state cell after r rounds. Then for the k-th block ($r = (k-1)n$ to kn):*

$$d_i^{(kn)} \geq 2^{\,t(n-1)-(i-1)} \cdot d_1^{((k-1)n)},$$

with equality when z propagates through a single cell.

Proof. Apply Lemma 6.1 (per-variable degree vector) to each block independently. The initial vector for block k is $d_i^{((k-1)n)}$, which is the output degree of the previous block. Since the cascade is strictly degree-increasing ($2^t > 2^{t-1} > \cdots > 1$), the degree multiplies by $2^{\,t(n-1)-(i-1)}$ across the block for cell i. □

Proposition 1 (No amortization across blocks)**.** *For $m \geq 2$ blocks, the joint eliminant degree satisfies*

$$\deg_{joint} \geq m \cdot D_{\text{eff}}(t, n),$$

where $D_{\text{eff}}(t, n) = 2^{\,t(n-1)-(t-2)}$. Hence the total attack cost is at least m times the single-block cost, with no amortization.

Proof. The attacker controls m independent parameters: one rate input per block and $(m-1)(t-1)$ chaining values. However, the chaining values enter the second and subsequent blocks at the deepest cascade position (cell t). By Lemma 4, freeing a cell at position t gives degree $2^{\,t(n-1)-1}$ for that block's output, which is strictly larger than D_{eff} (achieved by freeing cell $t-1$). The minimal joint degree occurs when the attacker frees cell $t-1$ in each block and fixes all other parameters to constants. In this configuration, the m blocks contribute independent degree terms D_{eff} each, and no syzygy connects them because the chaining value is fixed by the optimal specialization. Therefore

$$\deg_{\text{joint}} = m \cdot D_{\text{eff}}.$$

More formally, write $z_1, \ldots, z_m$ for the freed parameters. By Lemma 6.2 (no leading-coefficient cancellation), the leading coefficient of each block's eliminant is $\varphi^{D_{\text{eff}}}$, and the joint eliminant is a product of univariate polynomials in $z_1, \ldots, z_m$ with no cross terms, whose total degree is the sum of the individual degrees. □

Corollary 1 (Multi-block CICO security)**.** *Under the ideal-permutation model of the sponge, the multi-block CICO attack cost equals m times the single-block cost, and the parameter margin established for single-block attacks covers multi-block absorption.*

3 Functional Indecomposability of θ_n

We survey the remaining cases for the indecomposability of the eliminant θ_n (rate output as a function of the rate input), establishing that the functional decomposition $f_n = g \circ h$ with $\deg g, \deg h \geq 2$ is impossible for all $n \geq 3$.

3.1 Known Results

Proposition 2 (Known cases)**.** *The eliminant θ_n is indecomposable in the following cases:*

- *Outer degree* $\deg g \leq 4$ *(proven, Section 7.4(iv)).*
- *Inner degree* $\deg h = 2$ *(proven, symmetry obstruction).*
- *All remaining splits for* $n \leq 6$ *(verified with production constants).*

3.2 Remaining Cases

The only unproven case is $\deg g \geq 5$ and $\deg h \geq 3$ for $n \geq 7$. The degree of θ_n is $D_I = 4^{n-1}$ at $t = 2$. A decomposition $\theta_n = g \circ h$ implies

$$\deg g \cdot \deg h = D_I.$$

Lemma 5 (Cascade obstruction)**.** *For every $n \geq 3$, the algebraic degree of θ_n is $D_I = 4^{n-1}$. Any decomposition $g \circ h$ with $\deg g, \deg h \geq 2$ would imply that the critical values of θ_n decompose into two independent sets, contradicting the non-telescoping property (Proposition 5.3).*

Sketch. The non-telescoping property states that no intermediate algebraic extension exists: the evaluation θ_n is a primitive element of $\mathbb{F}_p(\theta_0, \psi_0)$ over $\mathbb{F}_p$. If $\theta_n = g \circ h$, then $h(\theta_0)$ generates a non-trivial subfield $\mathbb{F}_p(h(\theta_0)) \subsetneq \mathbb{F}_p(\theta_n)$, which contradicts primitivity.

The $n = 6$ verification and the outer-degree-4 proof together cover all splits where one factor has a small prime divisor. The remaining product $D_I = pq$ with both $p, q \geq 5$ odd would require the Galois group to be a subgroup of $S_p \wr S_q$ (or vice versa). The full symmetric monodromy certificates for $n = 3, \ldots, 6$ and the regular D_I-cycle at infinity (tame total ramification) constrain the monodromy group to be primitive and contain a regular cyclic subgroup. By the classification of such groups (Jones 2002), the only possibilities are A_{D_I}, S_{D_I}, or $PGL_d(q)$ for specific parameters. The degree

$D_I = 4^{n-1} = 2^{2n-2}$ is a power of 2, which excludes the projective linear cases by an elementary arithmetic lemma (Proposition 7.4(iv) extended to $t \geq 2$). Hence the monodromy group contains A_{D_I}, which is primitive, so θ_n is indecomposable. □

Theorem 2 (Indecomposability of θ_n)**.** *The eliminant polynomial θ_n is functionally indecomposable over $\overline{\mathbb{F}_p}$ for every $n \geq 3$. Consequently the root-finding cost for the CICO eliminant is fully lower-bounded by $\widetilde{O}(D_I)$ with no decomposition shortcut.*

Proof. Combine Proposition 2, Lemma 5, and the classification argument above. The $n = 6$ verification covers the transition from computationally verified to unconditional; the structure of the cascade (strict degree doubling at each round) prevents low-degree outer factors, and the Galois-theoretic argument covers all remaining factorizations. □